BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6475

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The moment of inertia of a ring is equal to:

(A) $1/12\ mr^2$ (B) mr^2

(C) $1/2\ mr^2$ (D) $2/5\ mr^2$

Q2. 1kWh = :

(A) 3.6μJ (B) 3.6mJ

(C) 3.6kJ (D) 3.6MJ

Q3. The rate of change of momentum of a body equals to:

(A) Work done (B) Applied power

(C) Applied force (D) Impulse

Q4. The range of projectile is maximum when projectile is thrown at an angle of:

(A) 30° (B) 45°

(C) 60° (D) 90°

Q5. If $\vec{A} = 2\hat{i} - \hat{j} + 2\hat{k}$ then A = :

(A) 2 (B) 3

(C) 5 (D) 9

Q6. The area of the parallelogram formed with $\vec{A}$ and $\vec{B}$ as two adjacent sides is equal to:

(A) AB sin θ (B) AB cos θ

(C) AB tan θ (D) AB

Q7. The units of gravitational constant have units:

(A) Nm^2kg^{-1} (B) $Nmkg^{-2}$

(C) Nm^2kg^2 (D) Nm^2kg^{-2}

Q8. Work have same dimensions as that of:

(A) Momentum (B) Power

(C) Torque (D) Impulse

Q9. Carnot engine consists of:

(A) Two steps (B) Three steps

(C) Four steps (D) Five steps

Q10. For adiabatic process, the first law of thermodynamics gives:

(A) Q = W (B) W = –ΔU

(C) Q = ΔU (D) Q = ΔU + W

Q11. If N is the number of rulings on the grating, then the resolving power in the mth order diffraction is equal to:

(A) R = N/m (B) R = m/N

(C) R = mN (D) R = 1/mN

Q12. The light from the Sun reaches the Earth with:

(A) Circular wave fronts (B) Plane wave fronts

(C) Spherical wave fronts (D) Elliptical wave fronts

Q13. The distance between a node and the next antinode is:

(A) 4λ (B) 2λ

(C) λ/4 (D) λ/2

Q14. The increase in the speed of sound for one degree Celsius rise in temperature by:

(A) $0.61cms^{-1}$ (B) $0.61ms^{-1}$

(C) $0.16cms^{-1}$ (D) $0.16ms^{-1}$

Q15. If the length of a simple pendulum is doubled, its period:

(A) Will not change
(B) Will also be doubled
(C) Will be halved
(D) Will increase by 1.4 times

Q16. A chimney works best when it is:

(A) Tall
(B) Wide
(C) Short
(D) Narrow

Q17. The SI unit of angular displacement is:

(A) Meter
(B) Degree
(C) Radian
(D) Revolution

Physics (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6475

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Show that the expression $v_i = v_i + at$ is dimensionally correct, where v_i is the initial velocity at t = 0, "a" is the acceleration and v_i is the velocity at time "t".

(ii) The wavelength λ of the wave depends on the speed v of the wave and frequency f. Decide which of the following is correct? $f = v\lambda$ or $f = v/\lambda$

(iii) Find the dimensions and hence the SI unit of coefficient of viscosity.

(iv) Does a dimensional analysis give any information on constant of proportionality that may appear in an algebraic expression? Explain.

(v) Define null vector. What is the difference between null vector and zero vector?

(vi) Write the position vector in one coordinate system, two coordinate system and three coordinate system.

(vii) If $\vec{A} = 2\hat{i} - 10\hat{j}$ and $\vec{B} = 6\hat{k}$. Find $\vec{A} \times \vec{B}$ (cross product).

(viii) Explain how the swing is produced in a fast moving cricket ball?

(ix) Show that the range of projectile is maximum when projectile is thrown at an angle of 45° with the horizontal.

(x) Describe a case when a massive body collide with light body at rest.

(xi) How the helmet safe from injury in accident?

(xii) Water flows out from a pipe at $3kgs^{-1}$ and its velocity changes from $5ms^{-1}$ to zero on striking the wall. Find the force exerted by the water.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) A girl drops a cup from a certain height which breaks into pieces. What energy changes are involved?

(ii) When rocket re-enters the atmosphere, its nosecone becomes very hot. Where does this energy come from?

(iii) Define absolute potential energy. Give its unit.

(iv) Define angular momentum. What is its direction?

(v) Define angular velocity and angular acceleration.

(vi) What is meant by moment of inertia? Explain its significance.

(vii) Under what conditions does the addition of two simple harmonic motions produce a simple harmonic motion also?

(viii) Describe some common phenomena in which resonance plays an important role.

(ix) Does the acceleration of simple harmonic oscillator remain constant? Is the acceleration zero anywhere?

(x) What is period of 250 cycles per second of sound waves?

(xi) Find the temperature at which the velocity of sound in air is two times its velocity at 0°C.

(xii) If velocity of sound is $332 ms^{-1}$ at 0°C then what will be its velocity at 10°C?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Under what conditions two or more sources of light behave as coherent sources?

(ii) How would you manage to get more order of spectra using a diffraction grating?

(iii) State Huygen's principle.

(iv) If a person was looking through a telescope at the full moon. How would the appearance of the moon be changed by covering half of the objective lens?

(v) Why is the astronomical telescope preferred to use in normal adjustment?

(vi) Why does the pressure of a gas in a car tyre increase when it is driven through some distance?

(vii) Is it possible to construct a heat engine that will not expel heat into the atmosphere?

(viii) State Carnot's theorem.

(ix) Differentiate between internal energy of a substance and internal energy of an ideal gas.

Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 01,04) Define scalar product of two vectors with examples. Write any four characteristics of scalar product.

Q5(b). (Marks 03) Two spherical balls of 2.0kg and 3.0kg masses are moving towards each other with velocities of 6.0 m/sec and 4 m/sec respectively. What must be the velocity of smaller ball after collision if velocity of bigger ball is 3.0 m/sec.

Q6(a). (Marks 05) Describe the stationary waves produced in a stretched string and prove that their frequencies are quantized.

Q6(b). (Marks 03) A 70kg man runs up a long flight of stairs in 4.0 s. The vertical height of the stairs is 4.5m. Calculate his power output in watts.

Q7(a). (Marks 05) Define laminar and turbulent flow and derive an expression for Bernoulli's equation.

Q7(b). (Marks 03) A 1000kg car travelling with a speed of $144kmh^{-1}$ rounds a curve of radius 100m. Find the necessary centripetal force.

Q8(a). (Marks 05) Define molar specific heats of a gas. Also show that $C_p - C_i = R$

Q8(b). (Marks 03) A block of mass 4kg is dropped from a height of 0.80m on to spring of spring constant $k = 1960Nm^{-1}$. Find the maximum distance through which the spring will be compressed.

Q9(a). (Marks 05) What is compound microscope? Describe its construction and working. Also calculate its magnifying power.

Q9(b). (Marks 03) Sodium light ($\lambda = 589nm$) is incident normally on a grating having 3000 lines per centimeter. What is the highest order of the spectrum obtained with this grating?

GROUP – II

Physics (Objective) Group - II

Time Allowed: 20 Minutes	**Maximum Marks:** 17	**Paper Code:** 6476

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If $\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$ then angle between $\vec{A}$ and $\vec{B}$ is:

(A) 0 (B) π/4

(C) π/2 (D) π

Q2. Dimensions of ratio of angular momentum to linear momentum is:

(A) $[M^{\circ}LT^{\circ}]$ (B) $[M^{1}L^{1}T^{1}]$

(C) $[M^{1}L^{2}T^{1}]$ (D) $[M^{-1}L^{-1}T^{1}]$

Q3. Dimensions of $\sqrt{(F\ell/m)}$ are:

(A) $[M^{\circ}LT^{-1}]$ (B) $[ML^{-1}T]$

(C) $[ML^{2}T^{-3}]$ (D) $[ML^{-1}T^{-1}]$

Q4. Magnifying power of telescope is:

(A) f_e/f_o (B) f_o/f_e

(C) $f_e f_o$ (D) $1/(f_o f_e)$

Q5. In Michelson interferometer a fring is shifted each time the mirror is displaced through:

(A) λ (B) λ/2

(C) λ/4 (D) Zero

Q6. If pendulum vibrate with frequency 0.5Hz, then its length will be:

(A) 10cm (B) 50cm

(C) 80cm (D) 99cm

Q7. Bernoulli's equation is based upon law of conservation of:

(A) Momentum (B) Energy

(C) Mass (D) Charge

Q8. If speed of moving body is doubled its K.E. is:

(A) Doubled (B) Halved

(C) Unchanged (D) 4 times

Q9. SI unit of molar specific heat is:

(A) $Jmol^{-1}K^{-1}$ (B) $J\ mol\ K^{-1}$

(C) J mol K (D) $J\ mol^{-1}$

Q10. Highest efficiency of heat engine whose lower temperature is 17°C and higher temperature is 200°C is:

(A) 70% (B) 100%

(C) 35% (D) 38%

Q11. The stretched string of length 2m vibrates in 2 segments. The distance between two consecutive nodes is:

(A) 1m (B) 2m

(C) 0.5m (D) 4m

Q12. Tuning fork is source of:

(A) Heat (B) Light

(C) Sound (D) Electromagnetic waves

Q13. Rotational kinetic energy of the hoop moving down on inclined plane is:

(A) $1/2\ mv^2$ (B) mv^2

(C) $1/4\ mv^2$ (D) $3/4\ mv^2$

Q14. Pull of the Earth on 20kg body on surface of Earth is:

(A) 20N (B) 196N

(C) 19.6N (D) 1960N

Q15. Rate of change of momentum is called:

(A) Force (B) Pressure

(C) Tension (D) Impulse

Q16. Mass of fuel consumed by a typical rocket to overcome earth's gravity is:

(A) 1000kg/s (B) 100kg/s

(C) 10000kg/s (D) 10kg/s

Q17. $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k}$ is equal to:

(A) 0 (B) 1

(C) –1 (D) 2

Physics (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 6476

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Show that formula $T = 2\pi \sqrt{(\ell/g)}$ is dimensionally correct.

(ii) Add the following velocities given in m/s up to appropriate precision: 23.1 , 0.002 , 0.00023 , 5 – 12

(iii) Define the terms (a) Precision (b) Dimensions of physical quantities.

(iv) Write the dimensions of (a) Coefficient of viscosity η (b) Energy.

(v) Define the terms (a) Resultant vector (b) Subtraction of vector.

(vi) What is the unit vector in the direction of the vector $\vec{A} = 4\hat{i} - 3\hat{j}$?

(vii) Suppose the sides of a closed polygon represent vector arranged head to tail. What is the sum of these vectors?

(viii) Define the terms (a) The time of flight (b) The range of projectile.

(ix) What happened when light body collides with a massive body at rest?

(x) Find the time of flight of projectile when it is thrown at an angle of 30° with horizontal.

(xi) Explain the difference between laminar flow and turbulent flow.

(xii) Explain what do you understand by rocket motion?

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Calculate the work done in kilo joules in lifting a mass of 10kg through a vertical height of 10m.

(ii) A person holds a bag of groceries while standing still, talking a friend. A car is stationary with its engine running. From the stand point of work, how are these two situations similar?

(iii) Derive the mathematical expression for escape velocity.

(iv) What is meant by moment of inertia? Explain its significance.

(v) What is meant by angular momentum? Also define law of conservation of angular momentum.

(vi) Define angular acceleration. How angular and linear velocities are related? Explain.

(vii) What should be the length of a simple pendulum whose period is 1 second at a place where $g = 9.8ms^{-2}$.

(viii) If a mass spring system is hung vertically and set into oscillations, why does the motion eventually stop?

(ix) Describe two common phenomena in which resonance plays an important role.

(x) Is it possible for two identical waves travelling in the same direction along a string to give rise to a stationary wave? Explain.

(xi) Find the frequencies produced in organ pipe when it is open at both ends.

(xii) What are beats? Also mention one use of beats.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) In the Young's experiment, one of the slits is covered with blue filter and other with red filter. What would be the pattern of light intensity on the screen?

(ii) Differentiate the interference and diffraction patterns of light.

(iii) Write the conditions for detectable interference of light waves.

(iv) Why would it be advantageous to use blue light with a compound microscope?

(v) How convex lens act as a magnifying glass? Explain.

(vi) Give an example of a process in which no heat is transferred to or from the system but the temperature of the system changes.

(vii) Is it possible to convert internal energy into mechanical energy? Explain with an example.

(viii) Define internal energy of a substance. Is it state function?

(ix) How first law of thermodynamics explains human metabolism? Explain.

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 05) State and prove law of conservation of linear momentum.

Q5(b). (Marks 03) Two forces of magnitude 10N and 20N act on a body in directions making angle 30° and 60° with x-axis respectively. Find the resultant force.

Q6(a). (Marks 05) Explain the interconversion of potential energy and kinetic energy (i) When there is no frictional force (ii) When frictional force is present.

Q6(b). (Marks 03) The frequency of the note emitted by a stretched string is 300Hz. What will be the frequency of the note when the length of the wave is reduced by one third without changing the tension?

Q7(a). (Marks 05) State and prove Torricelli's Theorem with diagram.

Q7(b). (Marks 03) Calculate the angular momentum of a star of mass 2.0×10^{30}kg and radius 7.0×10^{5}km, if it makes one complete rotation about its axis once in 20 days. What is its kinetic energy?

Q8(a). (Marks 05) Define simple pendulum. Show that its motion is SHM. Discuss its working derive relation for its time period.

Q8(b). (Marks 03) Estimate average speed of nitrogen molecules in air under standard conditions of pressure and temperature.

Q9(a). (Marks 02,02,01) Explain the diffraction of X-rays by crystal and derive Bragg's law. What are the uses of diffraction of X-rays.

Q9(b). (Marks 03) A simple astronomical telescope in normal adjustment has an objective of focal length 100cm and eye piece of focal length 5.0cm. (i) Where is the final image formed? (ii) Calculate the angular magnification.